

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a1d1e3e63f077c874.jsonl`  
- SHA-256: `90dc81e422525462fffb172004befa580867c56b48b9a250e3985d7d45d00b54e`  
- Source modified: `2026-06-16T03:20:55+00:00`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:18.770Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/abs/2302.09657>

****Title:**** Table Tennis Stroke Detection and Recognition Using Ball Trajectory Data (2023)

****Found via:**** Shuttle/ball trajectory ? rally/point segmentation in racket sports search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:20.855Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:20:33.580Z)

Table Tennis Stroke Detection and Recognition Using Ball Trajectory Data

Basic Information

****Title:**** Table Tennis Stroke Detection and Recognition Using Ball Trajectory Data

****Authors:**** Kaustubh Milind Kulkarni, Rohan S Jamadagni, Jeffrey Aaron Paul, Sucheth Shenoy

****Submission Date:**** February 19, 2023

****arXiv ID:**** 2302.09657

Methods and Approach

The work employs a multi-stage pipeline for stroke analysis:

****Ball Tracking:**** Two object detection approaches were implemented and benchmarked? "YOLOv4, a traditional object detection model, and TrackNetv2, a temporal heatmap based model."

****Stroke Segmentation:**** The authors developed a mathematical method to extract temporal boundaries. They explain that "a mathematical approach developed to extract temporal boundaries of strokes using the ball trajectory data yielded a total of 2023 valid strokes."

****Stroke Classification:**** The system uses "a temporal convolutional network developed [that] performed stroke recognition on completely unseen data." Additionally, the authors note that "several machine learning and deep learning based model architectures have been trained for stroke recognition."

Key Results

- ****Accuracy on unseen data:**** 87.155%
- ****Dataset size:**** 2,023 valid strokes from four professional players across six stroke classes
- ****Additional capabilities:**** The system successfully "detecting services and missed strokes"

Dataset and Novel Contribution

The dataset was captured using "a single camera setup positioned in the umpire's view." The work's innovation lies in demonstrating that "meaningful inferences such as stroke detection and recognition can be drawn using minimal input information"?specifically ball trajectory alone, rather than analyzing player body movements.

4. assistant (2026-06-16T03:20:36.561Z)

The fetch succeeded. This is a primary research paper (arXiv preprint) directly relevant to dimension 2 (shuttle/ball trajectory ? rally signal) and dimension 1 (boundary extraction). Let me extract the falsifiable claims.

5. user (2026-06-16T03:20:46.329Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:55.461Z)

Extracted and returned the structured output. Summary of what I found:

****Source:**** Kulkarni et al., "Table Tennis Stroke Detection and Recognition Using Ball Trajectory Data" (arXiv:2302.09657, submitted 2023-02-19). Quality: ****primary**** (arXiv research preprint).

****Most relevant findings to the research question (over-merge / segmentation gap, dimension 2):****

- A ****deterministic mathematical method**** extracts stroke temporal boundaries directly from ball trajectory ? yielding 2,023 valid strokes. This is a direct analogue to the project's missing boundary-split logic, and notably a ***non-learned*** approach, which fits the tiny-data constraint.
- Segmentation + recognition achieved from ****trajectory alone**** (no body pose), validating the 1-D-signal bet.
- Explicitly ****detects services and missed strokes**** ? directly bears on distinguishing in-play from serves/dead-time that bridge rallies into mega-windows.
- A TCN reaches ****87.155%**** recognition accuracy on unseen data (supports dimension-1 TCN candidates).
- Tracker-agnostic (validated on both YOLOv4 and ****TrackNetv2****), compatible with the project's

WASB/TrackNet stack.

Caveat: the WebFetch summary did not surface the paper's exact algorithmic detail of *how* the mathematical boundary extraction works (e.g., velocity-zero crossings, net-crossing logic), so the precise mechanism for splitting is unverified from this fetch and would need a full-text read to adopt.